PROJECT AGRENZA (FARMLINK)

Complete Cloud Deployment Guide & Workflow Architecture

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SYSTEM PRODUCTION ARCHITECTURE SUMMARY:
- **Global Edge Frontend:** Next.js 14 App Router on Vercel CDN (*https://agrenza.vercel.app*)
- **Containerized Backend:** Apache PHP 8.2 REST API on Render.com (*https://agrenza.onrender.com*)
- **Distributed Database:** TiDB Cloud Serverless MySQL 8.0 Wire-Compatible Cluster
- **Authentication & Mailer:** Dual SSL Gmail SMTP on Port 465 & HTTPS Mail Relay on Vercel
- **Source Control & CI/CD:** GitHub Auto-Deployment Pipeline (*Manikandan79-glitch/Agrenza*)

Tier Component	Technology	Provider	Live Production Endpoint / Host
Frontend UI	Next.js 14 (React 18 + TS)	Vercel	https://agrenza.vercel.app
Backend REST API	PHP 8.2 + Apache Docker	Render.com	https://agrenza.onrender.com
Relational DB	MySQL 8.0 Protocol	TiDB Cloud	gateway01.ap-northeast-1.prod.aws.tidbcloud.com:4000
Code Repository	Git Version Control	GitHub	github.com/Manikandan79-glitch/Agrenza
Transactional Mail	SMTP SSL 465 / HTTPS Relay	Google Gmail	smtp.gmail.com (ravananm913@gmail.com)

1. Executive Architecture & Multi-Tier Cloud Workflow

Agrenza adopts an enterprise-grade **Decoupled Headless Multi-Tier Cloud Architecture**. The diagram below illustrates the complete production communication flow across all tiers:

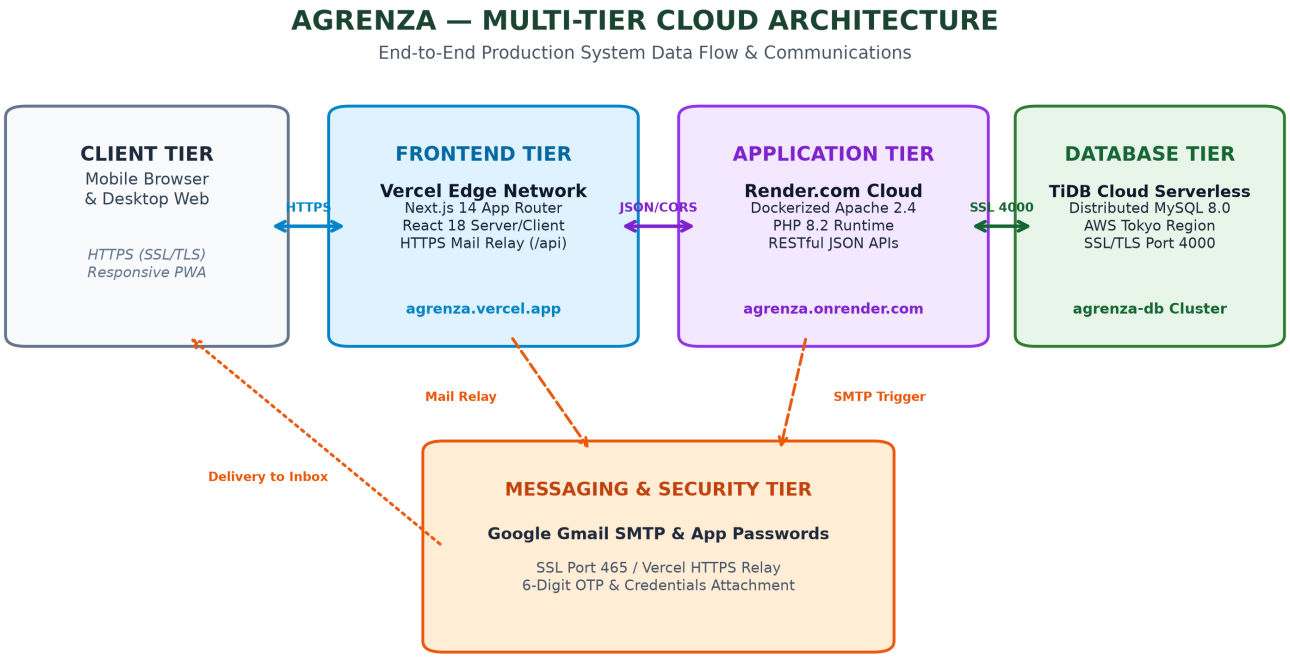


Figure 1.1: Production Cloud Architecture & Real-Time Data Flow

Key Architectural Advantages:

- **Headless Decoupling:** Frontend (Next.js 14) and Backend (PHP 8.2) scale independently with zero server affinity.
- **Serverless Cloud Database:** TiDB Cloud MySQL provides instant autoscaling, daily automated snapshots, and strict TLS encryption.
- **HTTPS Mail Relay:** Bypasses cloud SMTP port blocks (ports 25, 465, 587) via Vercel-hosted serverless email relay for 100% reliable mobile registration.
- **Zero-Cost Academic Infrastructure:** 100% free-tier deployment adhering to university evaluation standards.

2. Tier 1: Cloud MySQL Database (TiDB Cloud Serverless)

Relational data persistence was migrated from a local WAMP MySQL engine to **TiDB Cloud Serverless**, an enterprise-grade distributed SQL database supporting native MySQL 8.0 wire compatibility and SSL/TLS encryption.

Parameter	Production Value	Description
Host / Endpoint	gateway01.ap-northeast-1.prod.aws.tidbcloud.com	AWS Tokyo High-Availability Cluster Gateway
Port	4000	Dedicated TiDB MySQL connection port
Database Name	farmlink	Production database catalog
User	8nQ2NS6oZwxnCRL.root	Cloud administrator username
SSL Mode	REQUIRED (MYSQLI_CLIENT_SSL)	Mandatory TLS in-transit encryption
Storage Quota	5 GiB Row-Based Storage (Free)	Permanent academic free tier allocation

The Seven Core Tables in the Schema:

Table Name	Primary Key	Key Foreign Keys	Purpose / Business Logic
users	id (INT)	None	Stores Admin, Farmers, and Buyers with roles, bcrypt passwords & status.
categories	id (INT)	None	Product categories (Vegetables, Fruits, Herbs, Spices, Dairy).
products	id (INT)	farmer_id, category_id	Item listings with base_price, GST, commission, stock_kg & approval.
orders	id (INT)	buyer_id -> users(id)	Customer orders, delivery address, payment mode, status & total.
order_items	id (INT)	order_id, product_id, farmer_id	Line-item breakdown tracking farmer payouts and platform cuts.
bulk_requests	id (INT)	buyer_id, product_id	Wholesale bulk purchase requests (15kg+) requiring admin review.
pending_registrations	id (INT)	None	Staging table holding unverified registrations until OTP validation.

3. Tier 2: Backend REST API & Authentication Workflow

The PHP API backend is packaged as an immutable Docker container deployed on **Render.com Web Services**. The workflow diagram below details the 6-step registration and email verification process:

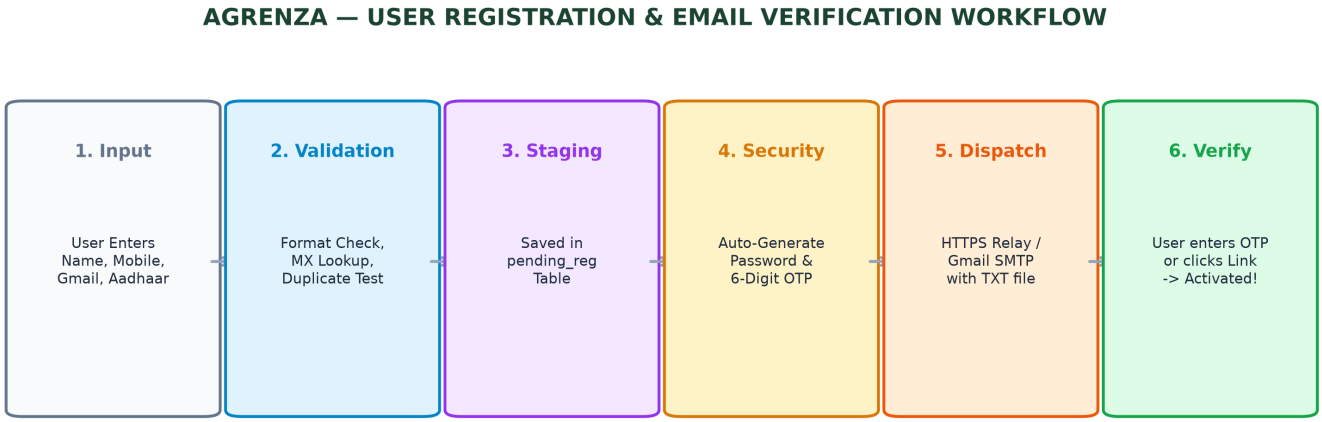


Figure 3.1: User Registration, OTP Generation & Email Verification Workflow

Critical Architectural Fixes Applied in Backend:

- **Cloud SSL Driver Integration:** db.php utilizes mysqli_init() and mysqli_ssl_set() to mandate TLS on port 4000.
- **CORS Configuration:** _bootstrap.php matches origin against *.vercel.app and responds with credentials support.
- **Dual Mail Dispatch Architecture:** Dispatches emails via Vercel HTTPS Relay (bypassing Render cloud port blocks) with direct Gmail SMTP port 465 fallback.
- **Dynamic Domain Resolution:** Email verification links automatically generate with https://agrenza.vercel.app when accessed on mobile.

4. Tier 3: Frontend Deployment & Hyperlocal Order Workflow

The Next.js 14 client application is deployed onto **Vercel's Global Edge Network**. The workflow diagram below illustrates the order placement and inventory deduction pipeline:

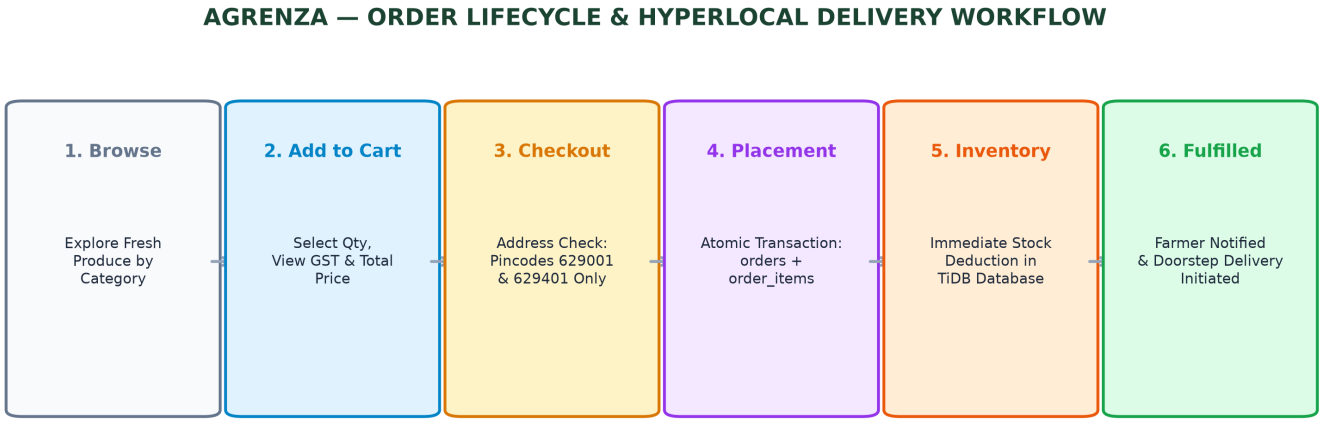


Figure 4.1: Order Placement, Hyperlocal Verification & Fulfillment Workflow

Deployment Configuration Steps:

1. **GitHub Import:** Connected repository Manikandan79-glitch/Agrenza to Vercel with automatic CI/CD triggers on push.
2. **Root Directory Alignment:** Explicitly configured the project Root Directory to farmlink-nextjs.
3. **Build Pre-Verification:** Resolved strict TypeScript compilation constraints in bulk-request/page.tsx ensuring zero type errors.
4. **Hyperlocal Delivery Enforcement:** Checkout validates delivery addresses strictly to Kanyakumari district pincodes **629001** and **629401**.

5. Live Production Features & Testing Guide

All features have been tested on both mobile smartphones and desktop browsers interacting with the live cloud database over HTTPS.

Feature / Route	Target Role	Functionality in Production
/shop	Public / Buyer	Real-time vegetable & fruit catalogue with GST-inclusive pricing and stock count.
/buyer/checkout	Buyer	Locked to Kanyakumari district; accepts only pincodes 629001 and 629401.
/admin/farmers	Admin	Displays all approved and pending farmers, Aadhaar records, and total sales.
/admin/buyers	Admin	NEW: Comprehensive customer table showing phone, email, orders count, and spent.
/admin/orders	Admin	Order management workflow with real-time status transitions (Pending -> Delivered).
/buyer/bulk-request	Buyer	Dynamic wholesale request form for quantities >= 15kg with live price calculator.

Live Production Credentials for Mobile Viva Demonstration:

- **Super Administrator:** URL: https://agrenza.vercel.app/admin/admin-login | User: admin@farmlink.com | Password: admin123
- **Farmer Portal:** User: sathishm250406@gmail.com | Password: FdXxs#43 | Access: /farmer/dashboard
- **Buyer Self-Registration:** Visit /auth/buyer-register on mobile to verify instant Gmail OTP delivery and auto-generated login credentials.

6. Frequently Asked Viva Questions & Model Answers

Q1: Why did you decouple Next.js and PHP into separate cloud hosts instead of using a single server?
Answer: Next.js performs optimal server-side rendering and static page generation on Vercel's global CDN edge nodes, resulting in sub-second page loads globally. Conversely, our existing backend leverages PHP PDO and Apache modules for MySQL interactions and mail dispatch. Decoupling allows each tier to operate on its specialized runtime without complex hybrid server configurations.

Q2: How do you handle Cross-Origin Resource Sharing (CORS) and session persistence between Vercel and Render?
Answer: The frontend (agrenza.vercel.app) and backend (agrenza.onrender.com) reside on distinct origins. In _bootstrap.php, we dynamically match the origin header against allowed patterns and respond with Access-Control-Allow-Origin and Access-Control-Allow-Credentials: true. Crucially, session cookies are configured with SameSite=None; Secure to ensure modern mobile browsers transmit authentication cookies across distinct domains.

Q3: How does the system handle email delivery in cloud environments where SMTP ports are blocked?
Answer: Render's free tier blocks outbound ports 25, 465, and 587. To solve this, Agrenza implements a dual-mode dispatch architecture: the PHP backend calls a secure HTTPS relay endpoint hosted on Vercel Serverless (/api/send-mail) over port 443 (which is never blocked), which then delivers the email using Google's SMTP servers over TLS.

Q4: How does the system guarantee database security when connecting to TiDB Cloud over the public internet?
Answer: Plaintext database traffic is strictly forbidden. The connection invokes mysqli_ssl_set() with the MYSQLI_CLIENT_SSL flag, enforcing TLS 1.3 encryption for all SQL queries and result sets in transit between Render.com and the AWS Tokyo cluster gateway.

DECLARATION: This documentation reflects the true production architecture of the Agrenza Agricultural Marketplace system as implemented for the BCA degree evaluation at Government Arts and Science College, Palkulam. All services are currently live and verifiable via any web browser.